



BATS SURVIVE WITH VIRUSES

Whenever we think about viruses, they are usually linked with illnesses, however in case of bats this scenario is a bit different as bats can coexist with the viruses without falling sick. They are known for being a potential host for many known and unknown viruses which can transmit to the human and animal populations. This coexistence of viruses within the bats is really significant in this time of frequent zoonotic disease outbreaks.

According to Dr. Wang Linfa from a research center in Singapore, the bats do not overfight to clear viruses, maybe that is why they can carry many viruses in them.

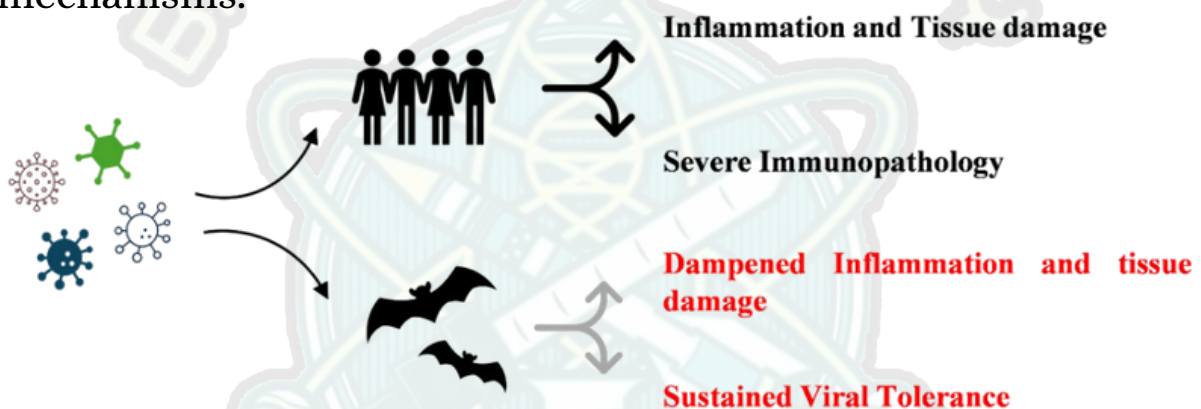




Banerjee et al. (2020) State that viruses such as filoviruses, coronaviruses, paramyxoviruses were found in healthy bats that are capable of causing severe diseases. However, these disease causing viruses do not harm the bat which is an interesting phenomenon scientists have long been intrigued upon. It is said that bat's body does not mount any destructive immune responses which is why they can coexist with all these deadly viruses. For example, some bat species can withstand viral infections with less clinical effects than humans when infected with viruses including coronaviruses and paramyxoviruses.

It should be mentioned that little is known about the regulation of viral infection in bats till this day. However, due to the frequent outbreaks and pandemics, understanding the relationship between host and the virus has caught a lot of attention.

After doing research it was found that, humans generate interferons (proteins that fight viruses) only when the body is infected with virus whereas bats continuously generate interferons which always give them a ready immunity. It is thought that this characteristic helps them to control viruses without causing inflammation which causes damage in the body. Furthermore, flight is also thought to be one of the reasons why bats can coexist with viruses. As flight most likely shapes their immune system as flight requires a high metabolic rate. High metabolic rate produces oxidative stress which can help the bats to evolve and regulate immune control mechanisms.



If further detailed study can be conducted on how bats can survive viruses this knowledge will help scientists to develop new strategies which will help to combat viral diseases in humans. Additionally, recognizing bats as a natural reservoir of viruses will further emphasize on preserving the natural habitat of the wild life. This will result in reducing the human and wildlife conflict. Also this awareness will prevent the risk of virus spillover to humans. Bats are really interesting creatures and studying them will unlock new insights in the field of medicine and immunology. Unveiling their secrets might give us the answer to future pandemics and help us to understand the immune tolerance.